FUNCTIONAL ANALYSIS OF THE CANCER GENOME AT NUCLEOTIDE-RESOLUTION

MAVES: Approaches, Analysis, and Interpretation – Nov 2025

Andrew Waters – Adams Lab

Atlas of Variant Effects

wellcome connecting science

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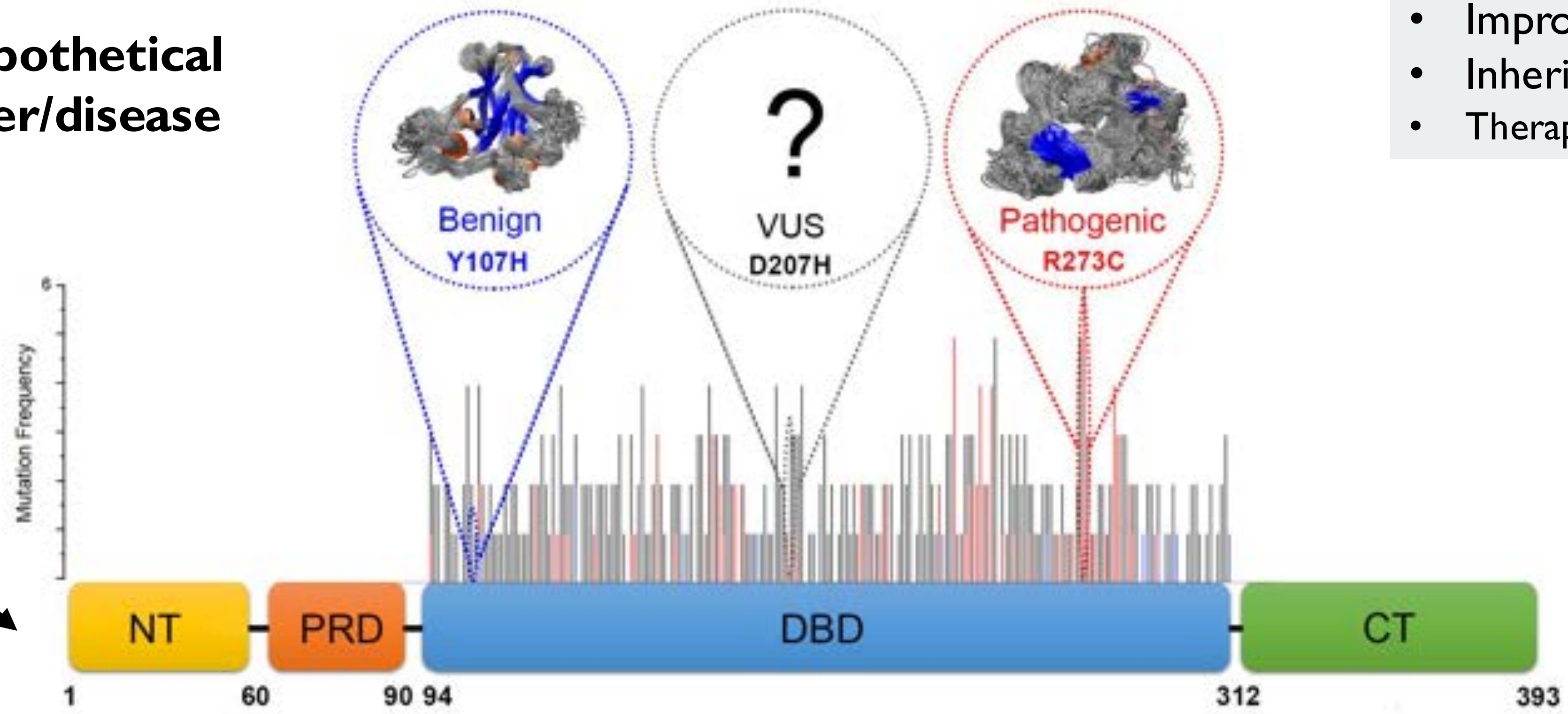
TYPES OF GENETIC VARIANT CLASSIFICATION

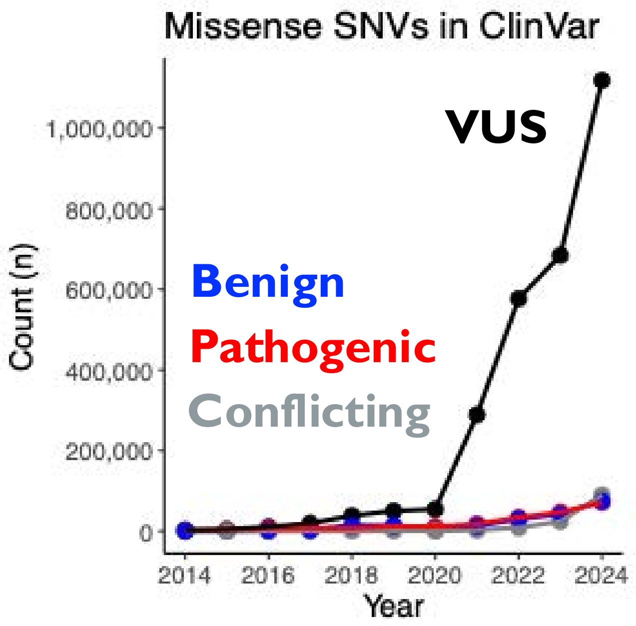
Finding and characterizing genetic variations associated with disease:

- Improved diagnosis/prognosis accuracy
- Inheritance biomarker/risk
- Therapeutic Development

Does nothing **NO IDEA** **Causes disease**

A hypothetical cancer/disease gene



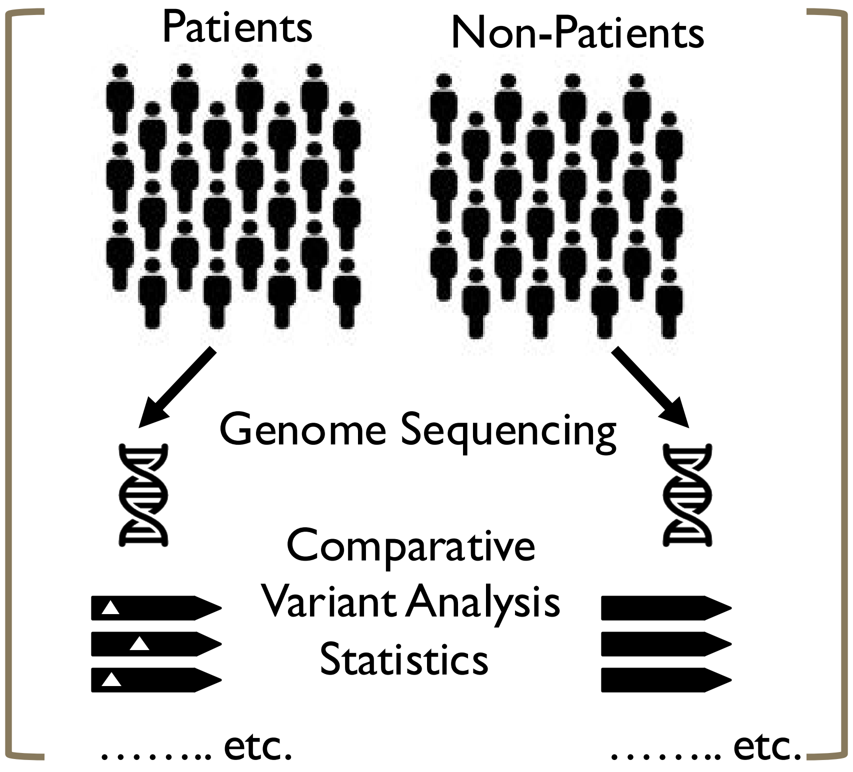


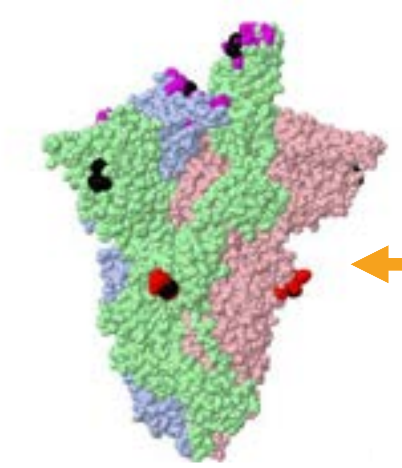
VUS are the most common type in our genomes

FUNCTIONAL ANALYSIS OF VARIANTS

Limitations of GWAS:

- Statistical power
- Finding **rare** variants
- **Partial penetrance** of variant phenotype





Variant function is not a solved problem

What do all variants in the protein/gene do?

What do polymorphisms do?

Where do we best target a small molecule?

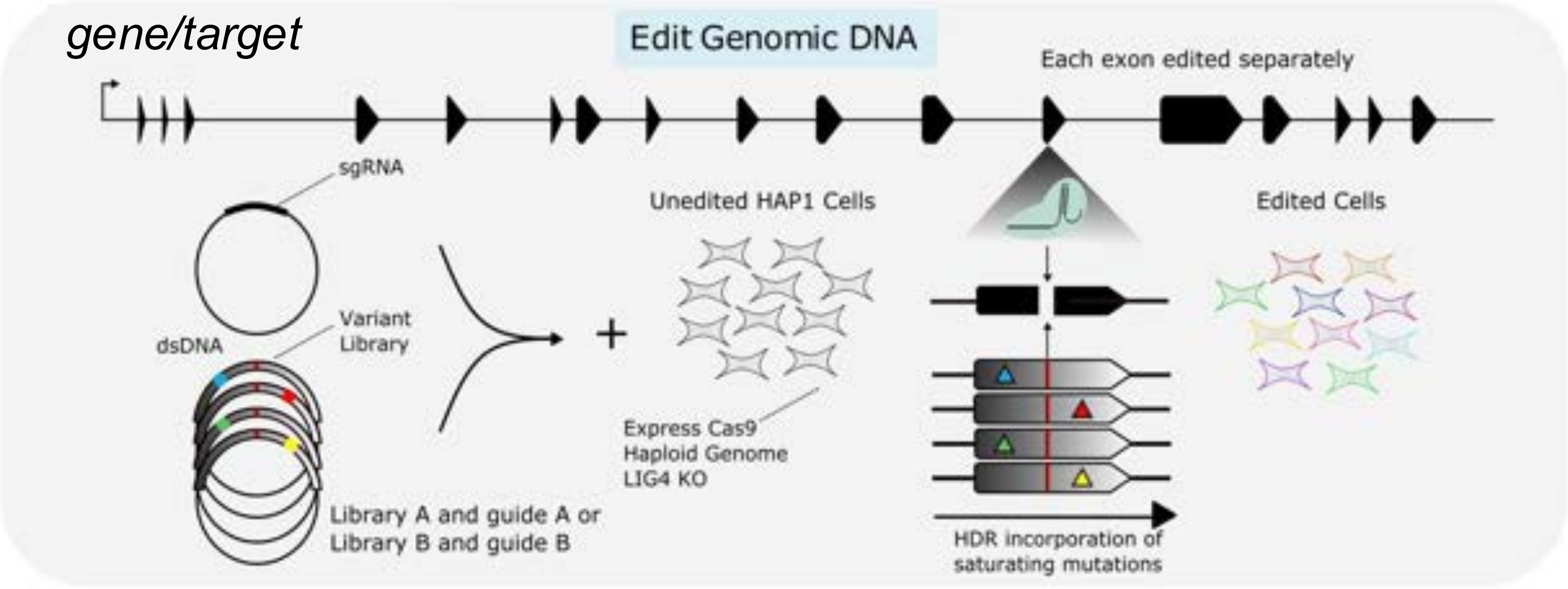
Functionally and **systematically** characterize **all** possible variants for certain genetic drivers of disease….

<u>Saturation Genome Editing</u> can do this

SATURATION GENOME EDITING

Variant Libraries are cloned from oligonucleotide pools

Many types of mutation are possible

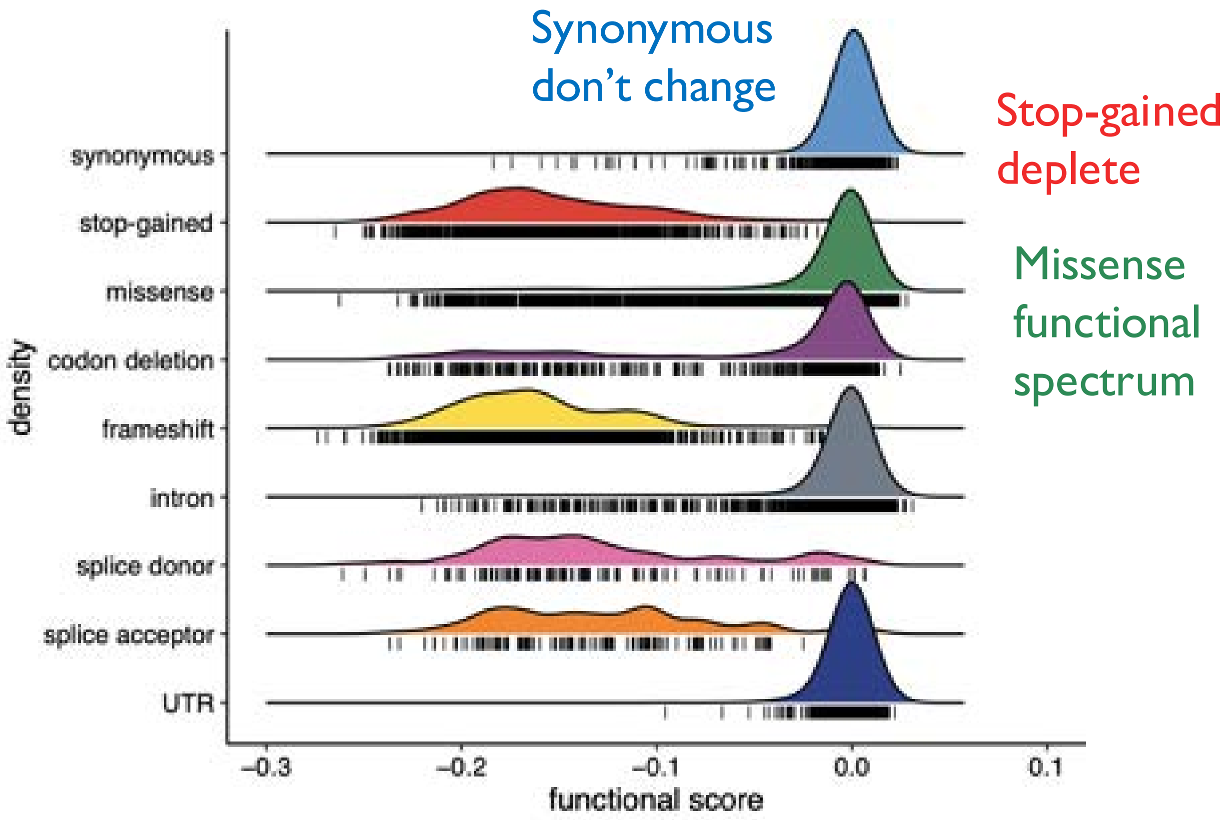


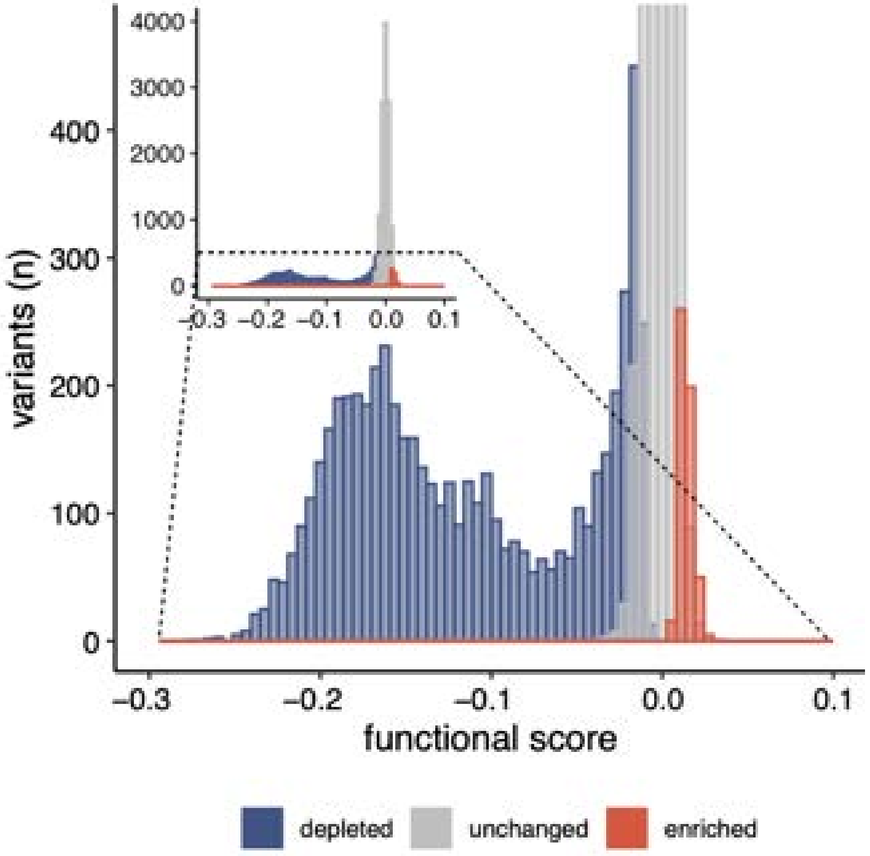
FOCUS ON SGE DATA – WHAT CAN IT TELL US?

18,108 variants
(6,196 disruptive)

BAP1- VARIANT DYNAMICS

Different dynamics for different mutational consequences

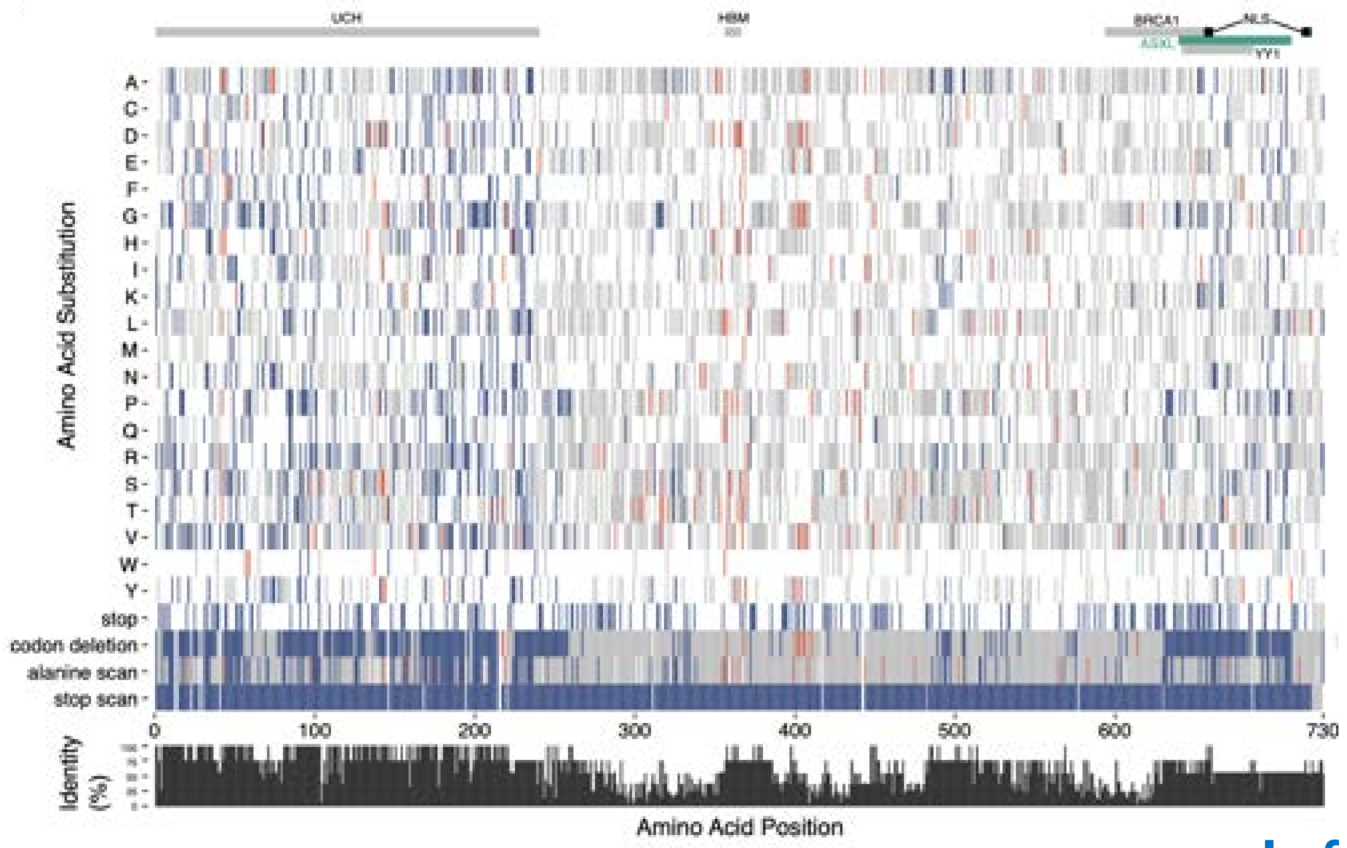




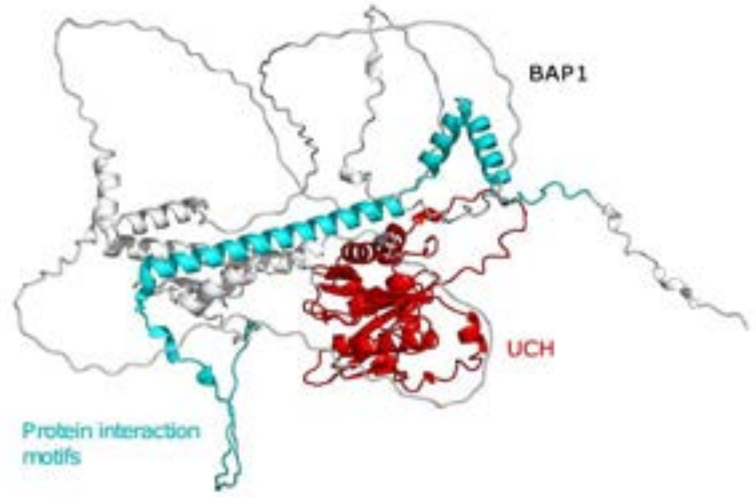
Variant pathogenicity classification:

Depleted: negative functional score and FDR <0.01
Enriched: positive functional score and FDR <0.01
Unchanged: FDR >0.01

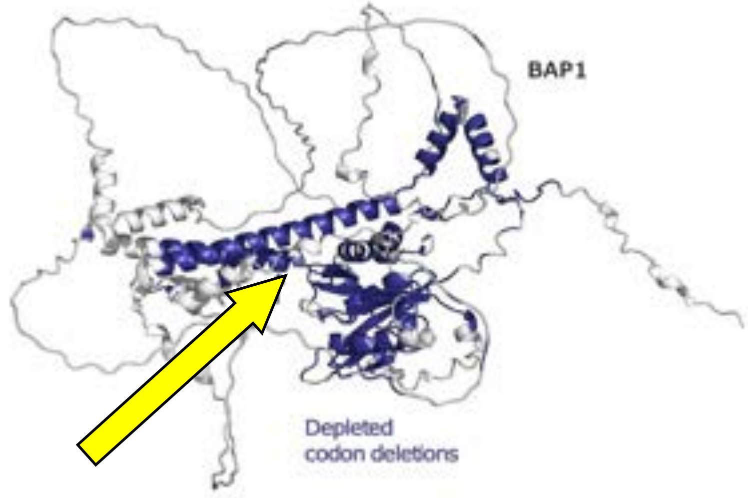
BAP1 – PROTEIN-LEVEL EFFECTS



Regions of intolerance correlated with high conservation

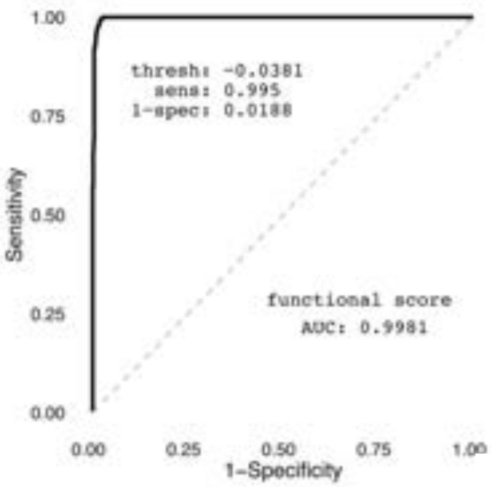


In-frame deletion of codons highlights known/novel domains



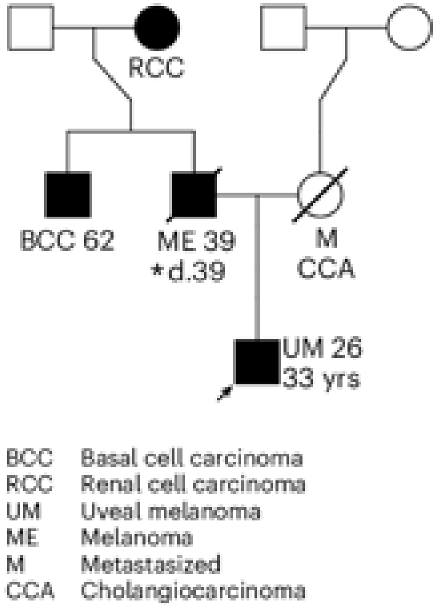
BAP1 - SGE CLINICAL APPLICATION

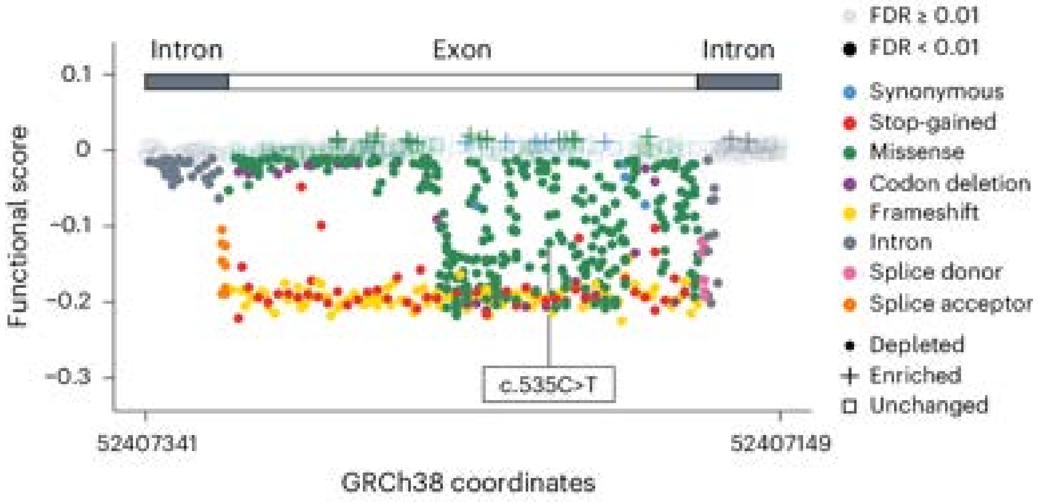
ROC analysis of SGE functional scores



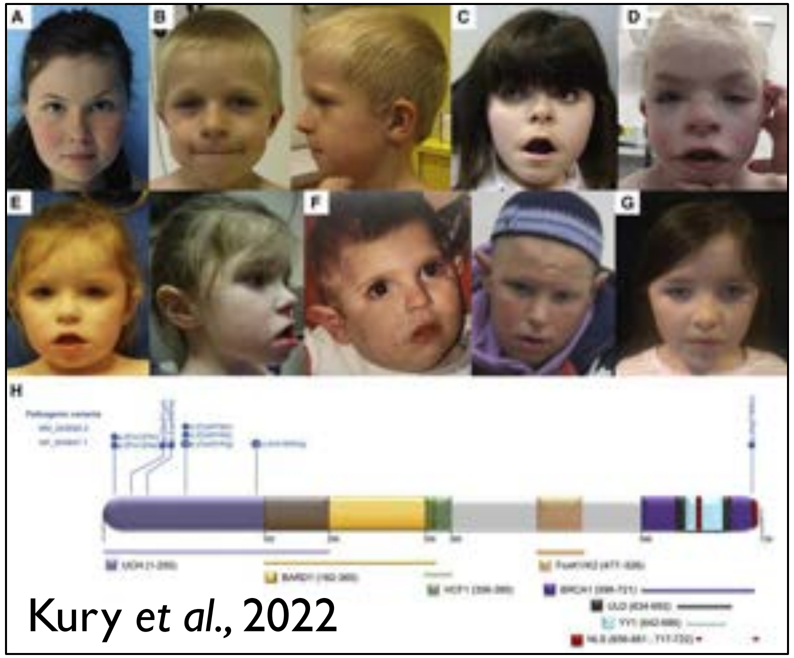
SGE scores classify known pathogenic and known benign variants separately, with almost zero error

Patient-level variant classification





Kürÿ–Isidor syndrome



Kury *et al.*, 2022

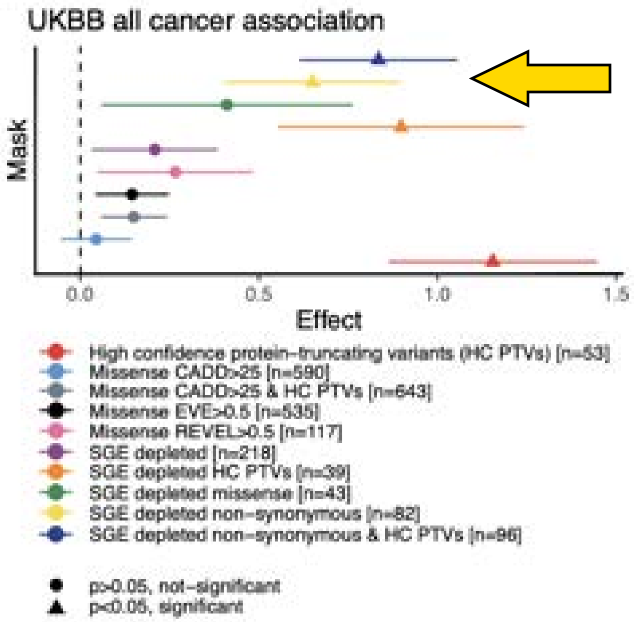
- Germline heterozygous *BAP1* variants
- Neurodevelopmental disorder
- **All SGE-depleted**

BAP1 – UKBB ASSOCIATIONS

Without SGE – not powered enough

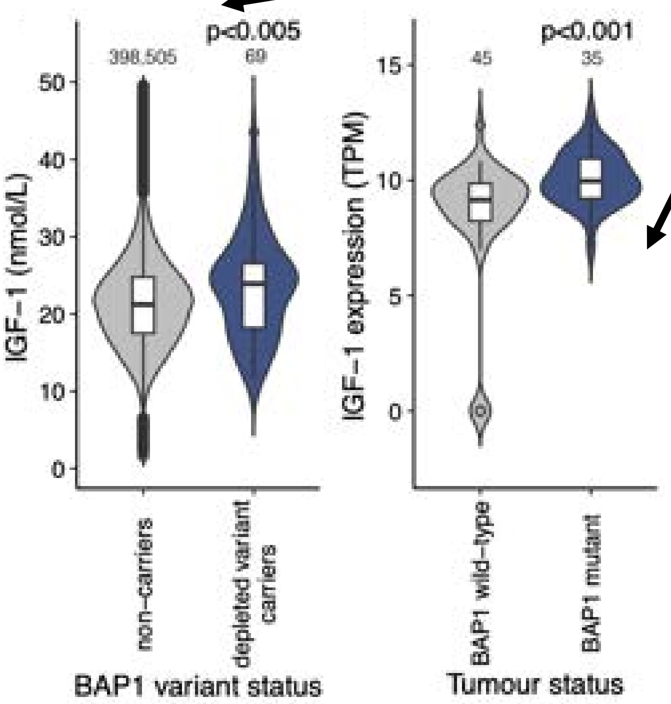
Phenome-wide association study with UK Bio Bank variants

- Non-synonymous SGE depleted variants significantly associated with cancer diagnosis



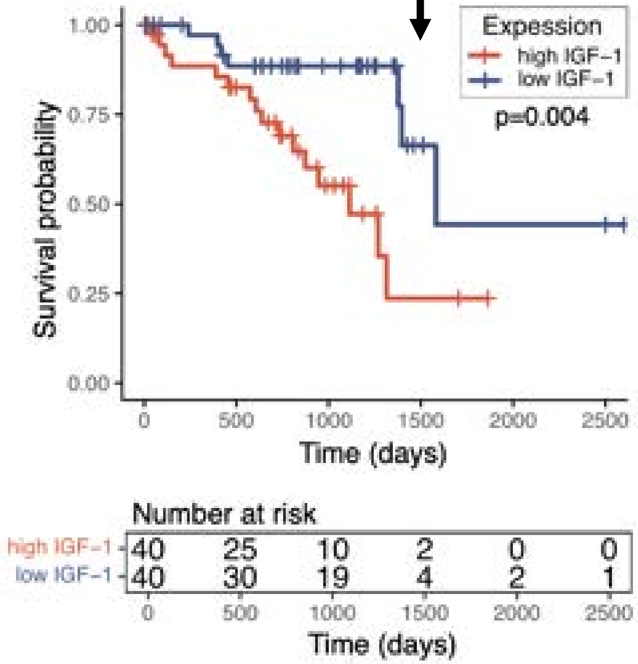
Dichotomous and Continuous traits

Circulating IGF-1 higher in *BAP1* SGE-depleted variant carriers



TCGA data, IGF-1 mRNA higher in uveal melanoma tumors

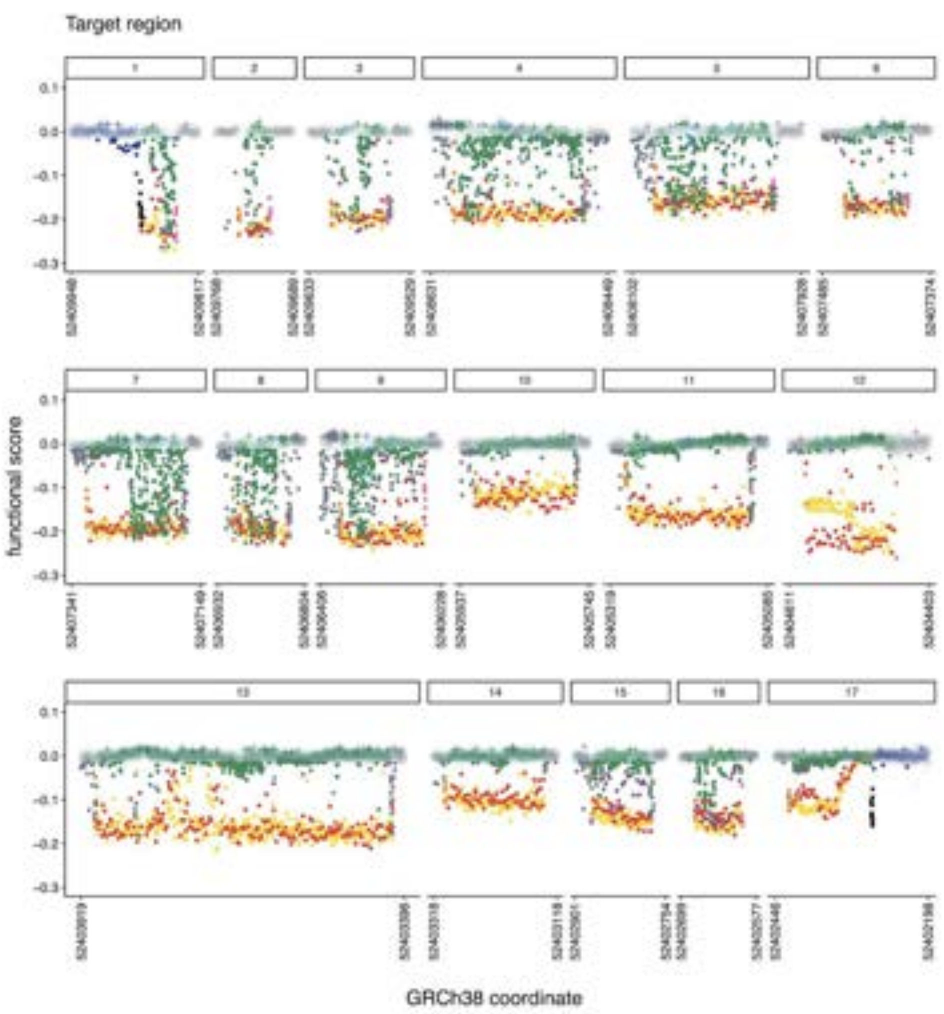
Higher survival in lower IGF-1 tumors



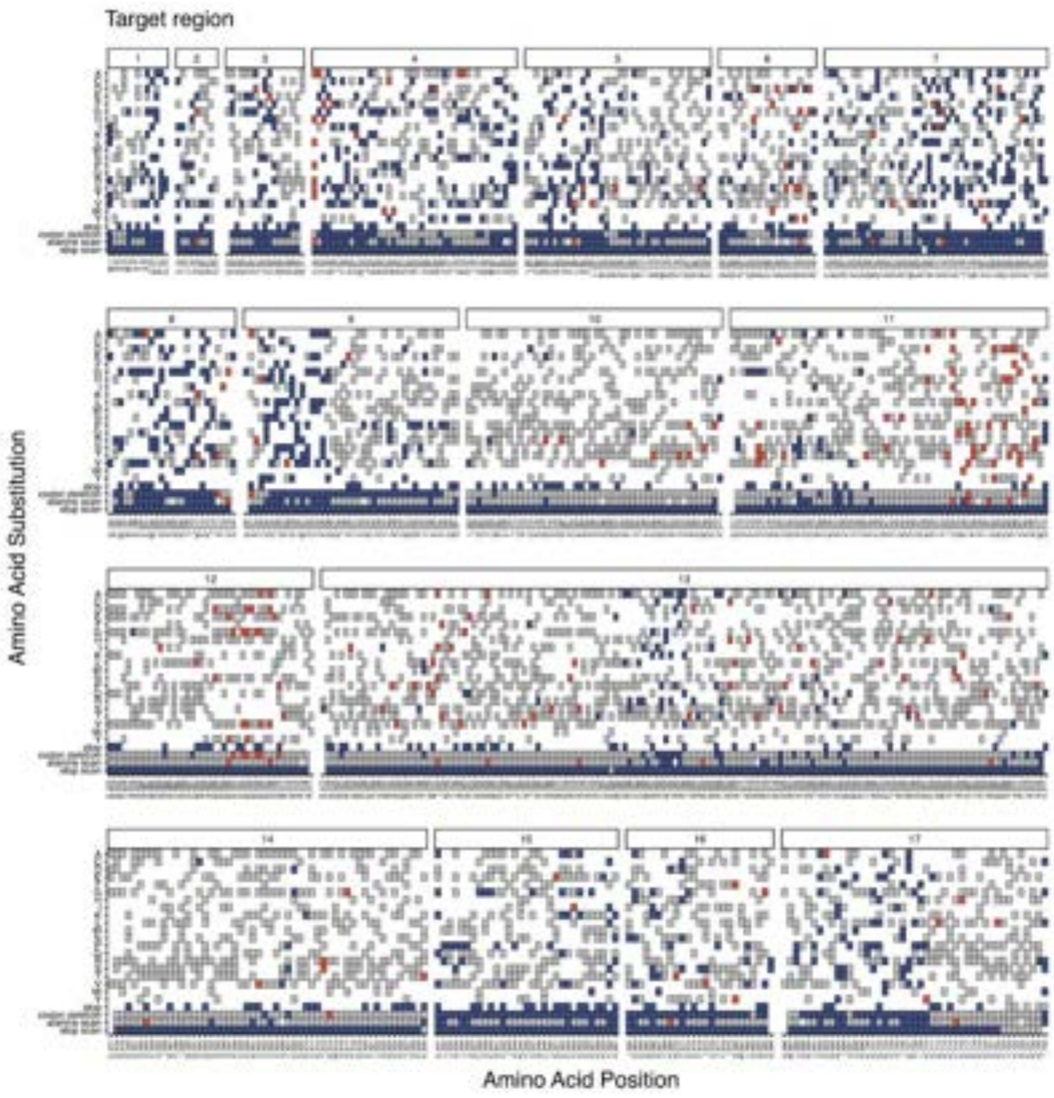
IGF-1 is a mitogen and fetal brain growth factor

BAP1 – COMPREHENSIVE MAPS

Nucleotide-resolution map



Amino Acid-resolution map



DO OTHER GENES HAVE SIMILAR VARIANT BEHAVIOURS?

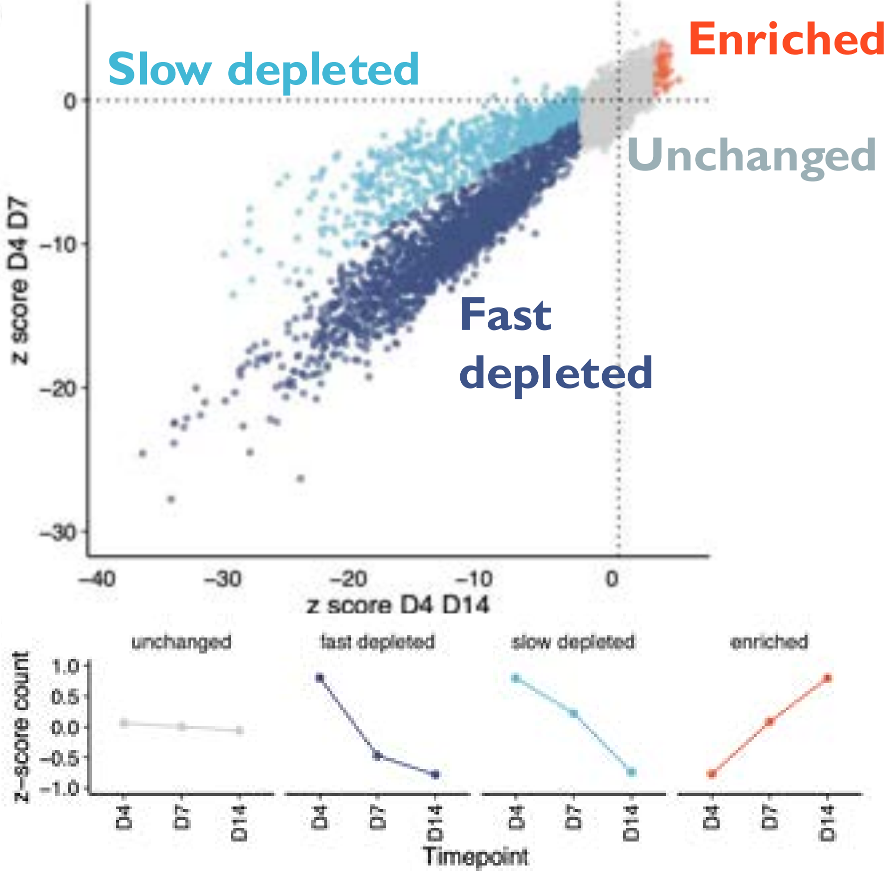
9,188 variants
(3,094 disruptive)

RAD51C – SPECTRUM OF EFFECT

Comparing **D4 vs D7** & **D4 vs D14**
allows us to examine variant kinetics

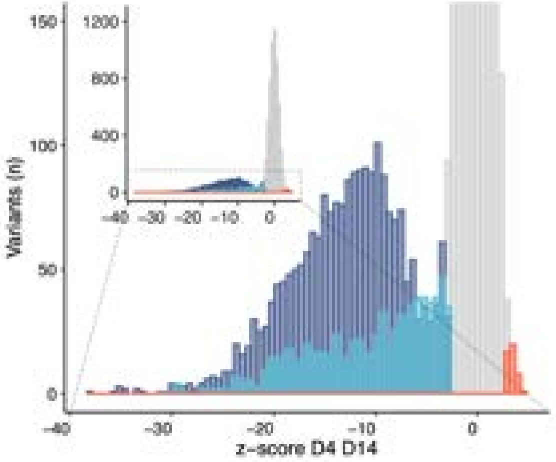
Classification of hypomorphic alleles

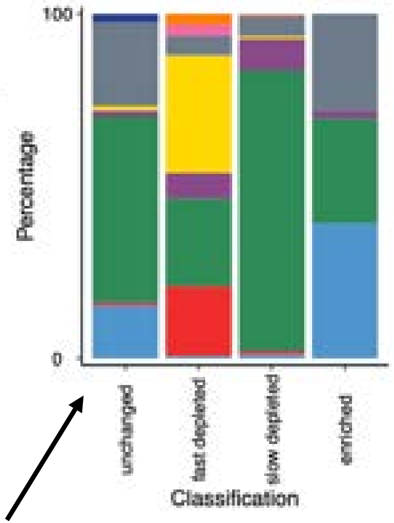
Gaussian Mixture Model on z-score comparison



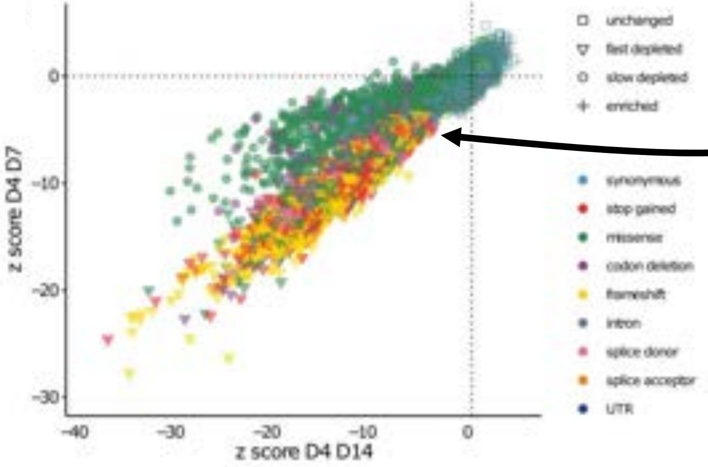
Slow depleted variant consequence

Slow, not necessarily less depletion





Slow-depletion variants are mostly missense



RAD51C – KINDRED RESOLUTION

Breast and Ovarian cancer

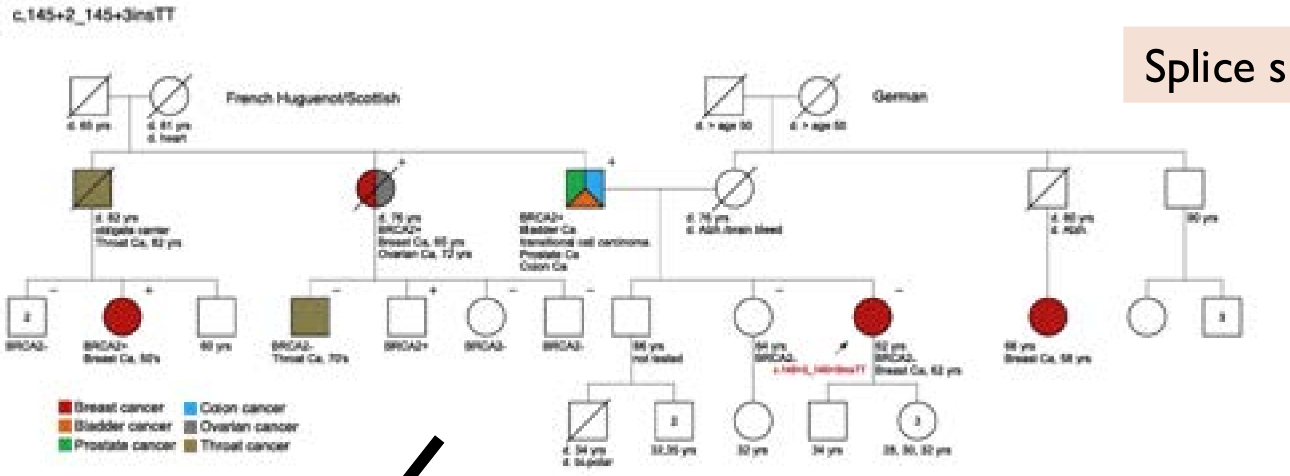
25 % HBOC = BRCA1/2

15% =

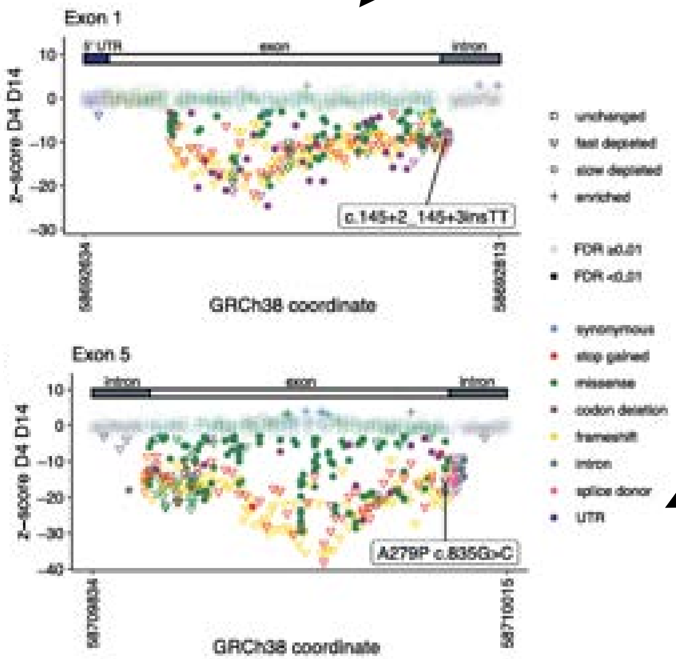
RAD51C/D
PALB2
BRIP1

Fanconi Aneamia

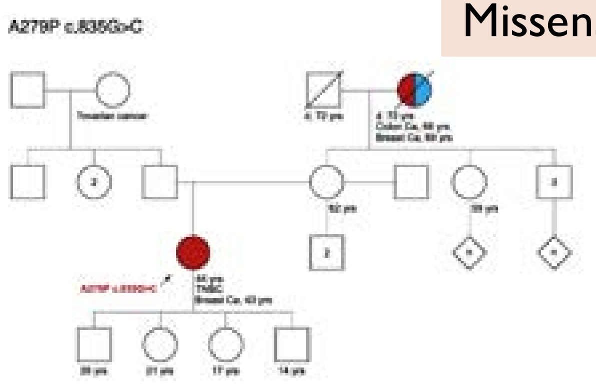
García-de-Teresa et al. (2020) *Genes.*



Splice site



Missense



Compromising variants reported

-families segregating *RAD51C* cancers

-important for clinical management

-family planning and genetic counselling

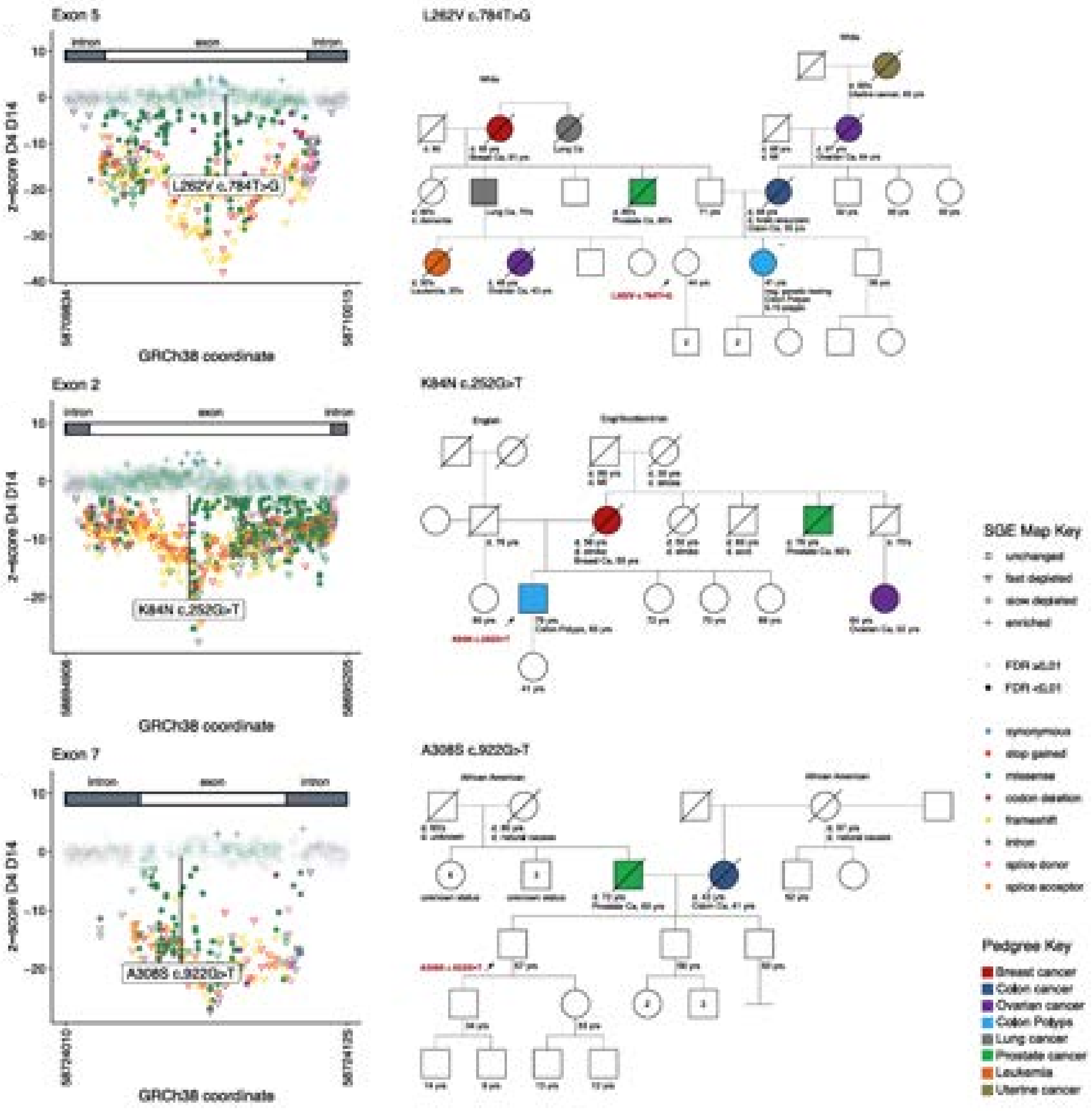
RAD51C – KINDRED RESOLUTION

Non-compromising variants also reported

-families segregating *RAD51C* cancers

-missense variants in probands are functional

-hugely important for diagnosis



All Missense

The trickiest class of VUS…

PAPERS FOR REFERENCE

Tim Brendler-Spaeth

Danielle Smith

Rebe Olvera-Leon

Fang Zhang

Dave Adams

Maria Jasin

nature genetics

Article

https://doi.org/10.1038/s41588-024-01799-3

Saturation genome editing of *BAP1* functionally classifies somatic and germline variants

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Check for updates

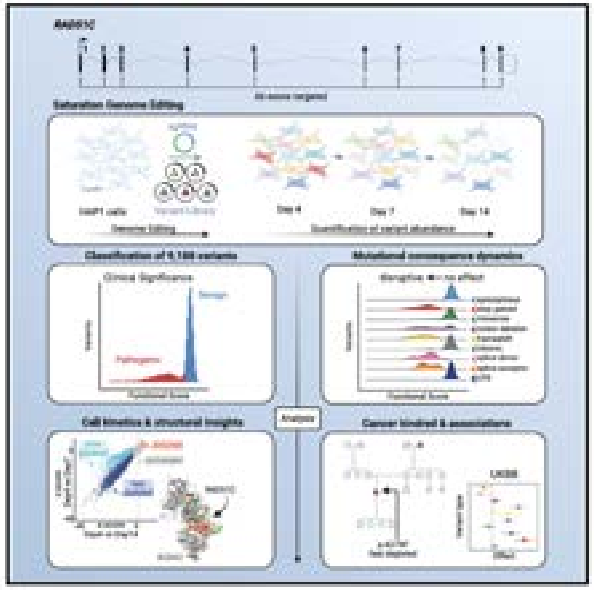
Andrew J. Waters[1], Timothy Brendler-Spaeth[1,2], Danielle Smith[1,2], Victoria Offord[1], Hong Kee Tan[1], Yajie Zhao[2], Sofia Obolenski[1], Maartje Nielsen[3], Remco van Doorn[4], Jo-Ellen Murphy[5], Prashant Gupta[1], Charlie F. Rowlands[6], Helen Hanson[7,8], Erwan Delage[1], Mark Thomas[1], Elizabeth J. Radford[1,9], Sebastian S. Gerety[1], Clare Turnbull[6,10,11], John R. B. Perry[2], Matthew E. Hurles[1] & David J. Adams[1]

Cell

Article

High-resolution functional mapping of *RAD51C* by saturation genome editing

Graphical abstract



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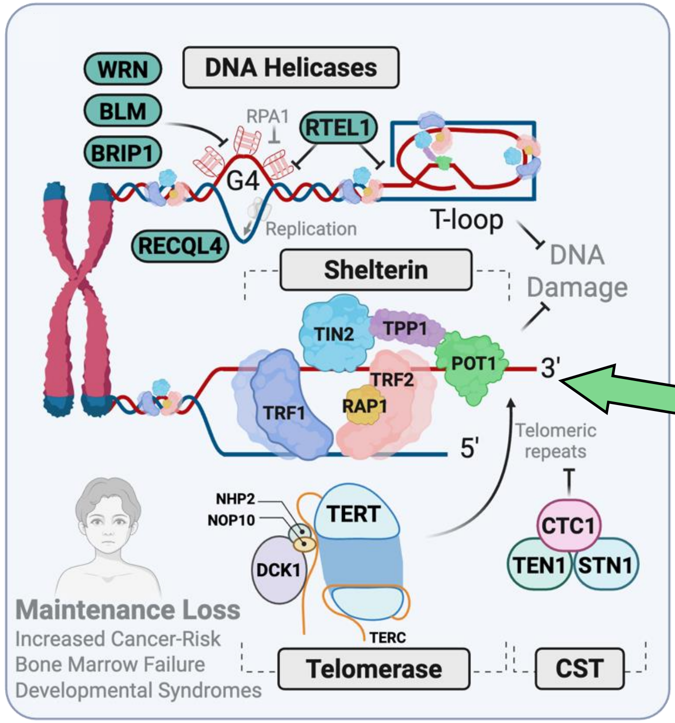
In brief

Functionally compromising variants in *RAD51C* pre-dispose individuals to breast and ovarian cancer; identifying these variants is critically important for the clinical management of patients and families. A saturation genome editing approach functionally classifies >99% of *RAD51C* coding sequence single-nucleotide variants and defines a spectrum of variant effects across multiple mutation types and exon/intron sequences.

Paper in revision…

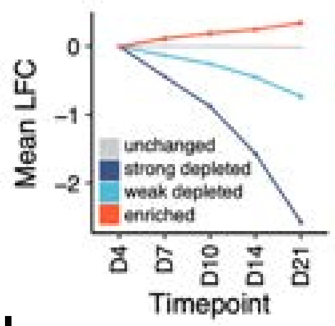
Sofia Obolenski

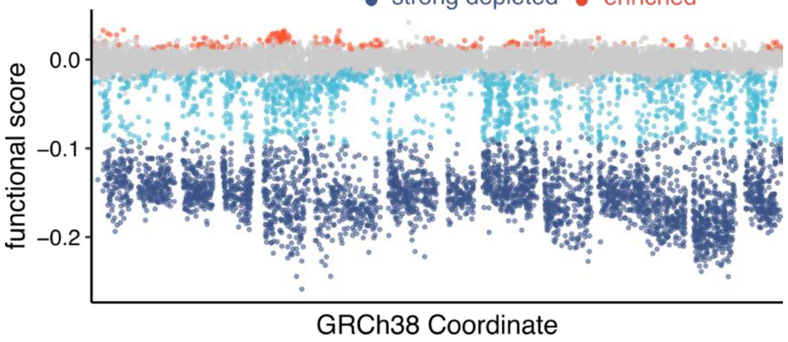
POT1 – SGE EFFECT SIZE & PENETRANCE



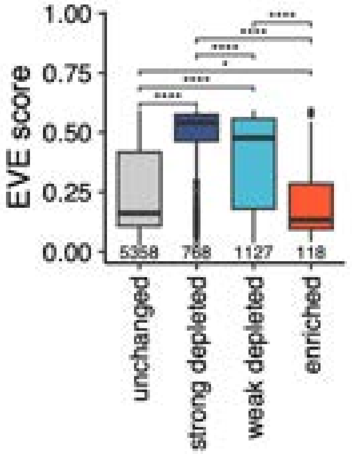
POT1

Weak vs strong depletion scored

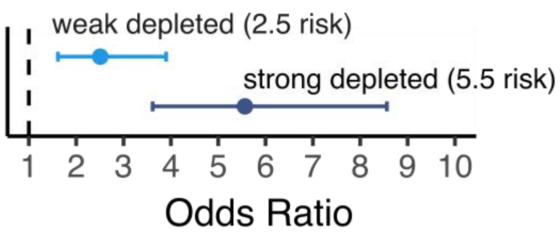


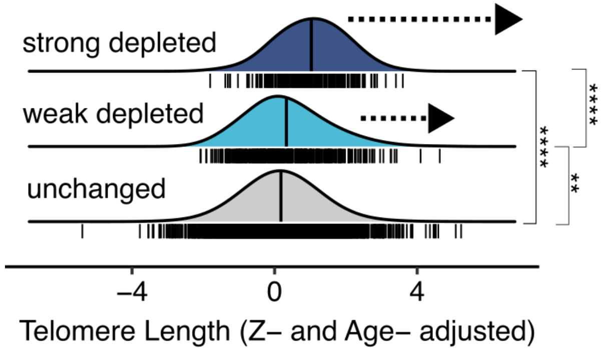


Conservation

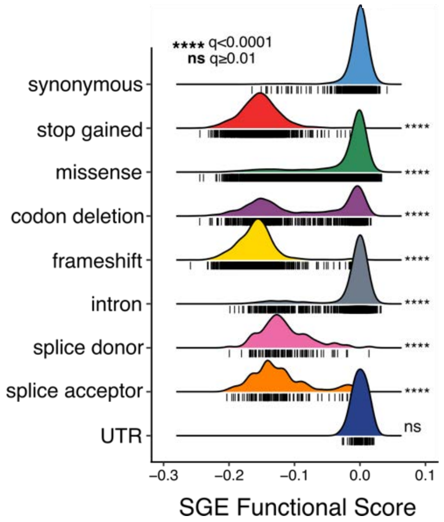


UKBB Cancer Risk

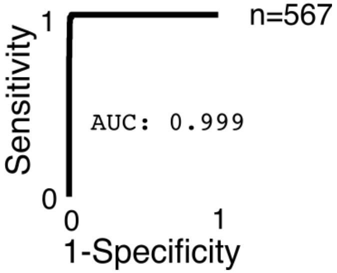


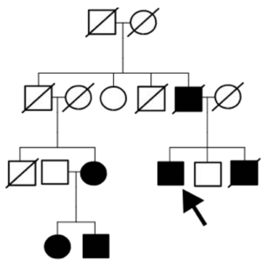


Telomere length increase more severe for strong depletion

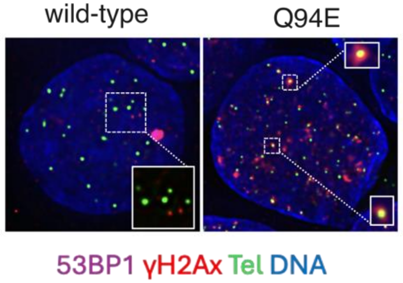


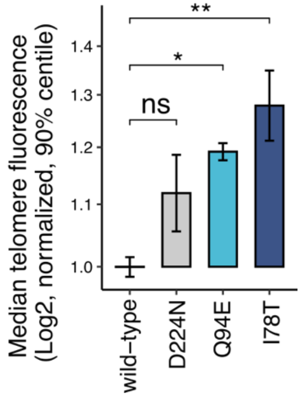
Rich clinical classification





Phenotypes validated



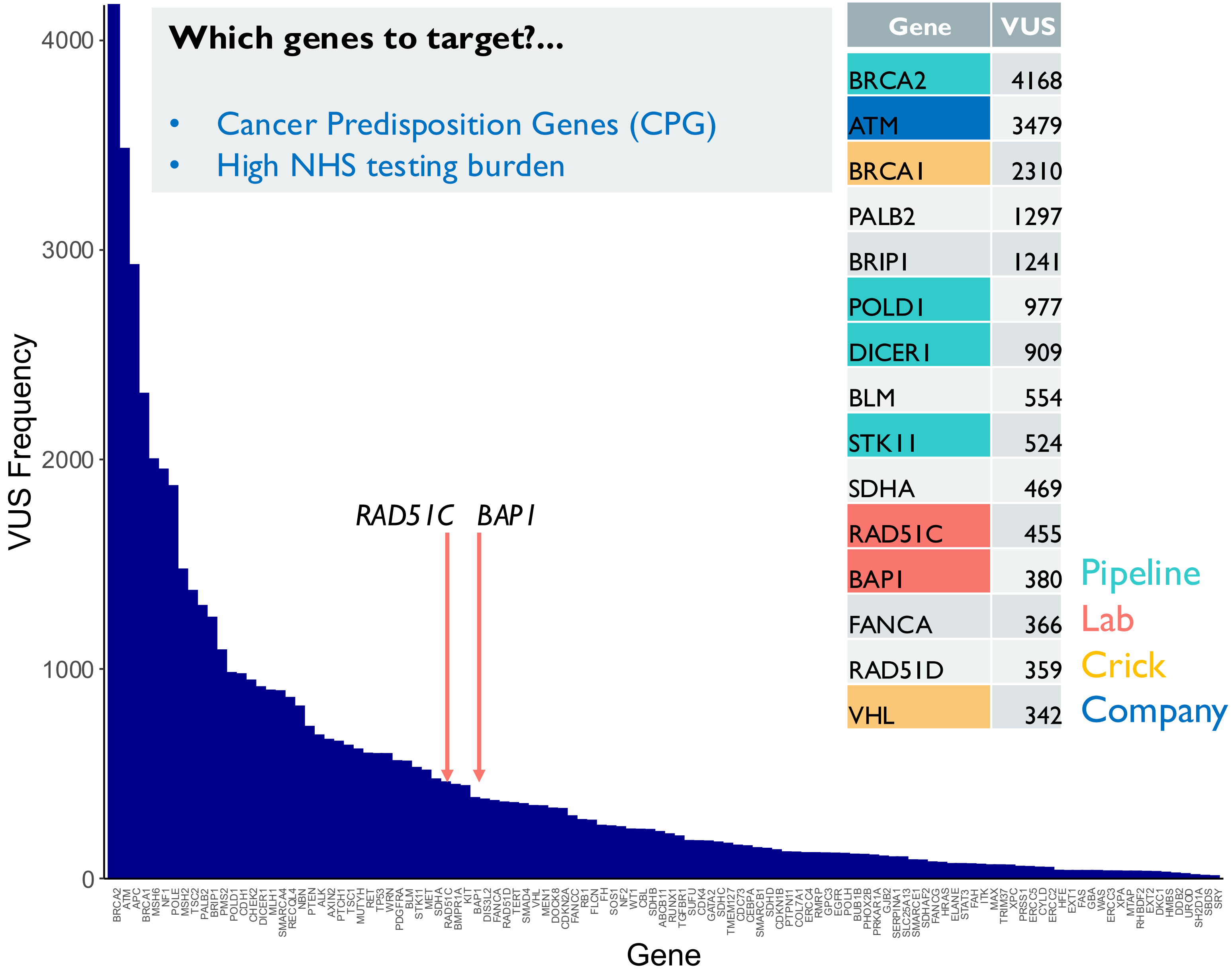


SANGER MAVE PIPELINE: SGE AT SCALE

Why scale?

- Many VUS
- Clinical management
- Protein function
- Drug discovery
- Fundamental change to genetics research

Design → Construction → Screening → DNA Extraction / Seq Library Prep → Data Analysis



Which genes to target?...

- Cancer Predisposition Genes (CPG)
- High NHS testing burden

Gene	VUS
BRCA2	4168
ATM	3479
BRCA1	2310
PALB2	1297
BRIP1	1241
POLD1	977
DICER1	909
BLM	554
STK11	524
SDHA	469
RAD51C	455
BAP1	380
FANCA	366
RAD51D	359
VHL	342

Pipeline
Lab
Crick
Company

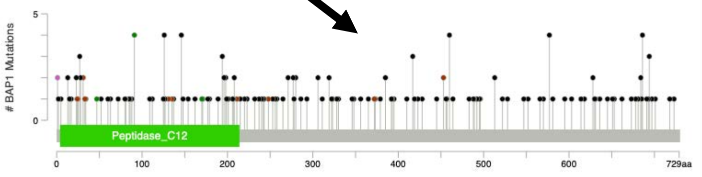
Atlas of Variant Effects

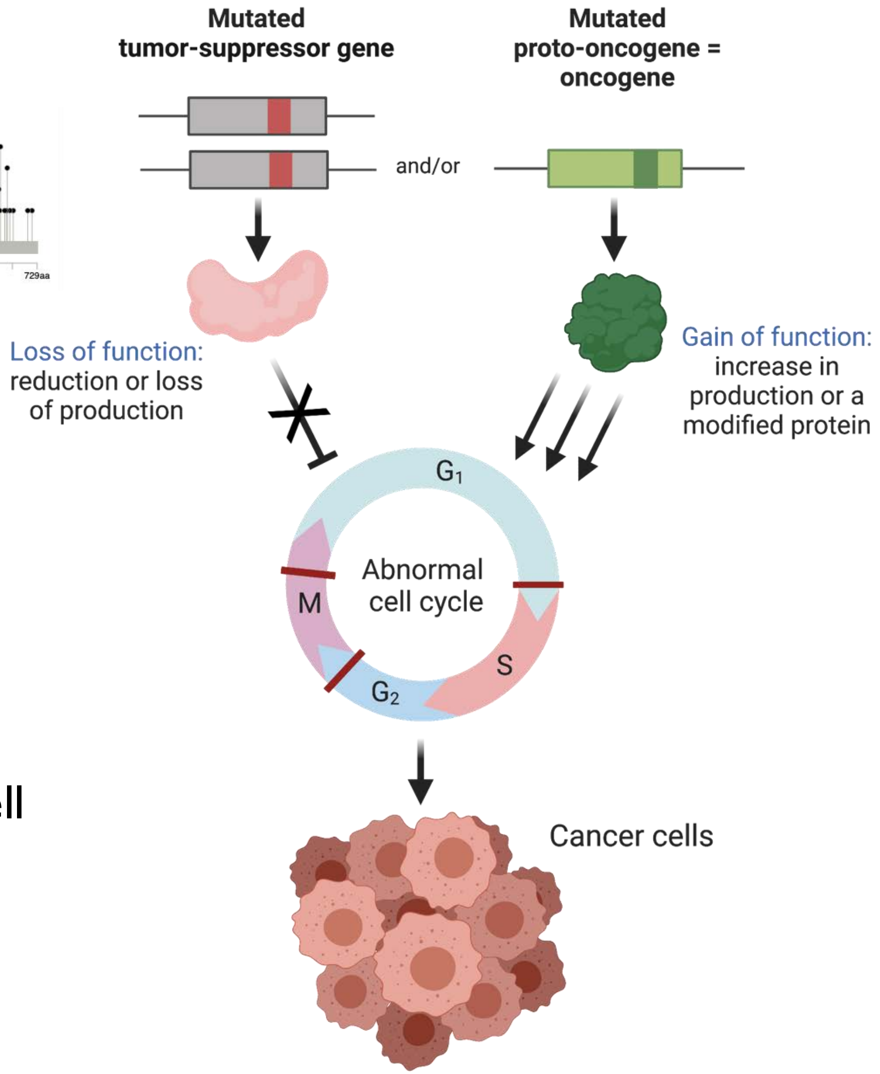
wellcome sanger institute

CLASSIFYING GAIN-OF-FUNCTION WITH SATURATION-SEQ

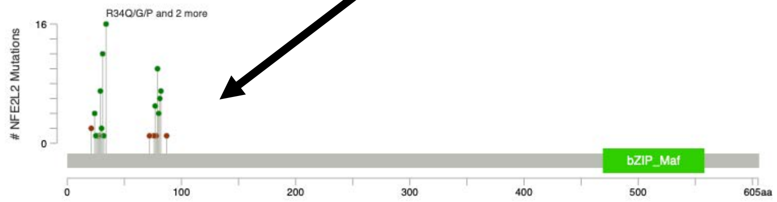
TUMOUR SUPPRESSOR GENES AND ONCOGENES

Variants through CDS





Variants at CDS hotspots



Tumour suppressor genes

- LoF
- Mutations occur throughout coding sequence
- Often essential when KO in cell models

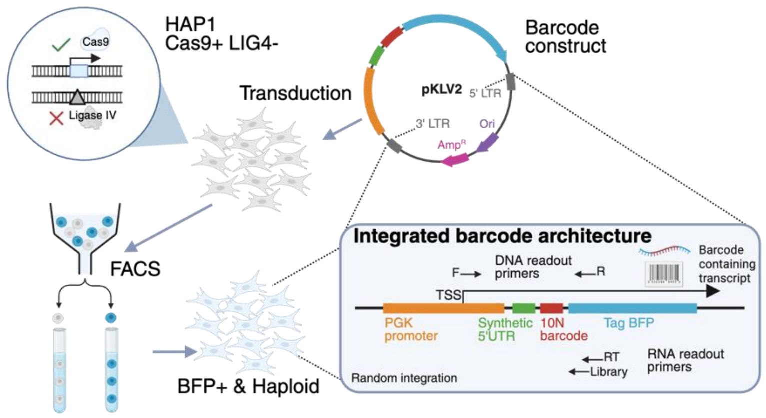
Oncogenes

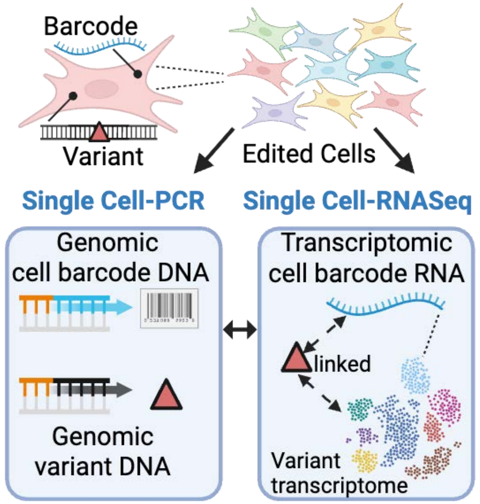
- GoF
- Occur at hotspots
- Often viable when KO in cell models

Need another SGE phenotyping approach!

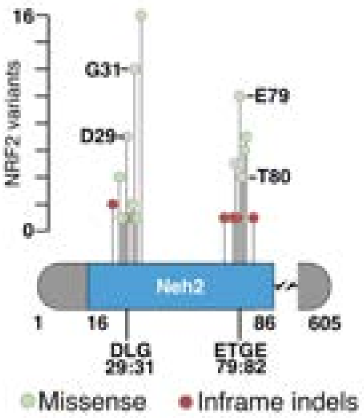
SATURATION-SEQ – MULTIPEXED GOF SCREENING

Experimental overview

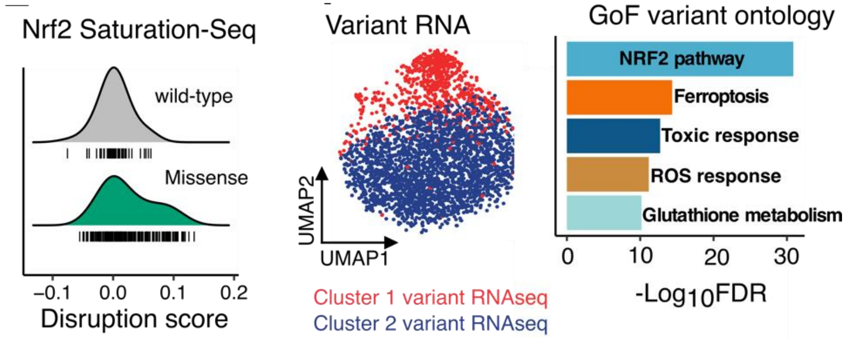




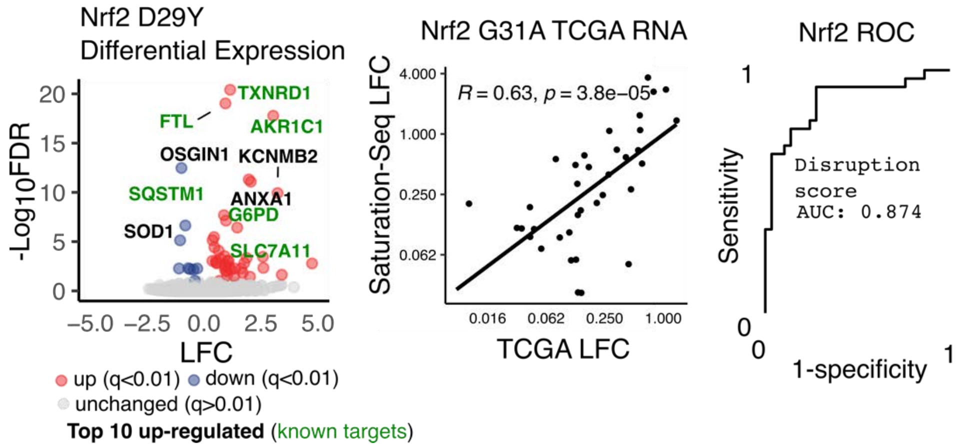
NRF2 redox protein
Recurrent GoF mutations



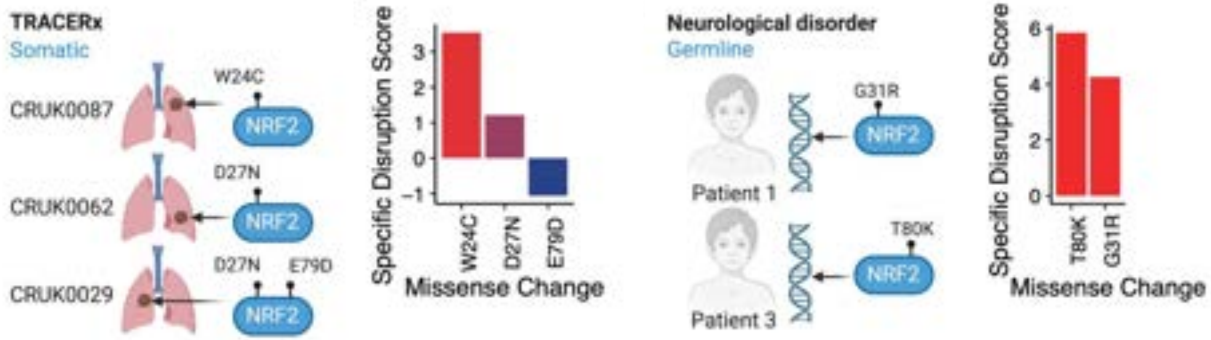
Missense variants form GoF cluster



High physiological and disease relevance



Somatic and Germline scoring



SUMMARY AND FUTURE DIRECTION

How we are using Saturation Genome Editing data now:

- VUS are a huge clinical burden that can be addressed by SGE
- We can use SGE to understand how genes work at high resolution
- SGE increases the statistical power of population-level associations - allows novel findings

How we can use SGE soon:

- Predict what regions of the genome are needed for health
- Begin to program DNA sequences to perform required biochemical functions?

THANK YOU

Dave Adams
Rebeca Olvera-Leon
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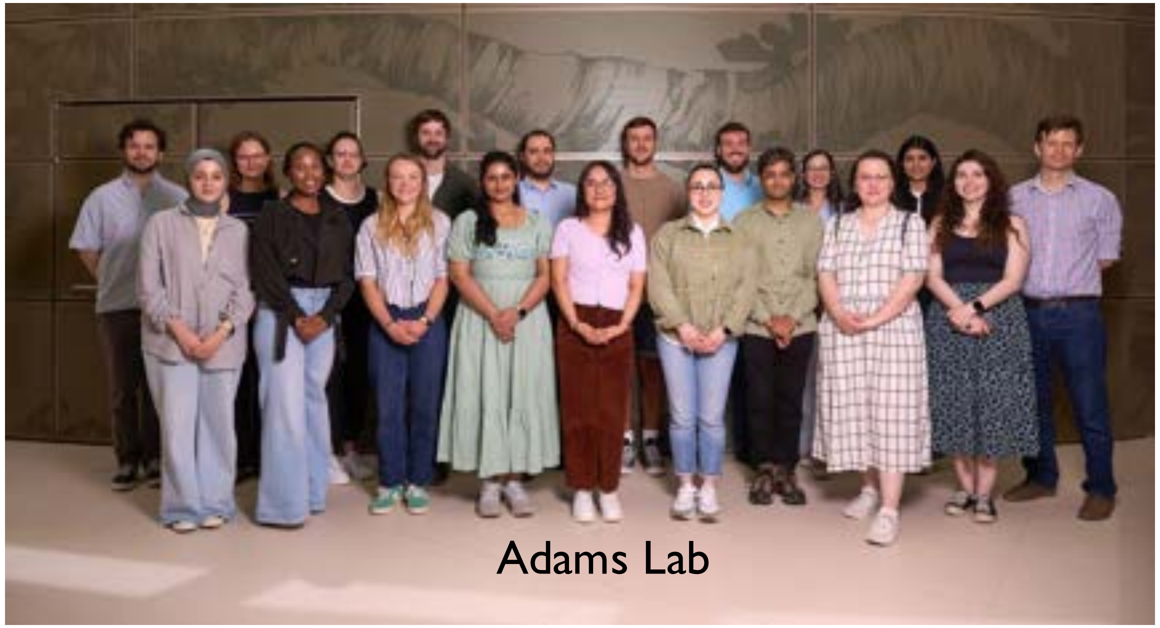
Magda Strauss

Hong Kee Tan
Sebastian Gerety
Lizzie Radford
Matt Hurles
Andrew Bassett
Victoria Offord
Tim Brendler-Spaeth
Danielle Smith
Chris Chan
Luca Barbon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Atlas of Variant Effects